Cyber Threat Intelligence (CTI) Strategy Report

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1. Objectives and Scope Definition

Define Objectives:
Document the specific goals of the CTI strategy, detailing what the organization aims to achieve (e.g., enhancing threat detection).

- Improve Ransomware Detection and Mitigation: Make early-stage Indicators of Compromise (IOCs) and Tactics, Techniques, and Procedures (TTPs) weaponized within the EDR environment to reduce dwell time of ransomware attacks.
- Secure Electronic Medical Records (EMRs) & Ensure Compliance: Keep patients' electronic medical records (EMRs) and telehealth platforms safe from unauthorized access to ensure zero breach integrity and meet strict HIPAA requirements.
- Enhance Incident Response Capabilities: Contain and recover clinical information faster in case of a security incident, maintaining clinical continuity and reducing disruption to patient care.

Define Scope:
Outline the areas within the organization that CTI efforts, including relevant departments, regions, or operational boundaries, will cover.

- Operational Systems: Production machines that run data for Electronic Medical Record (EMR) databases and external (public) Telehealth web/mobile applications.
- Departments included: IT Security, Cloud Operations, Compliance/Legal, Incident Response, and clinical staff endpoints.
- Infrastructure Boundaries: on-premises corporate endpoints, remotely located telehealth workstations, cloud-based SaaS applications, or critical backup environments (Veeam repositories).

Policy Alignment:
Identify key security policies aligned with the CTI goals. Briefly describe why these policies are priorities for this strategy and how they will influence subsequent phases.

- HIPAA Security Rule (Technical Safeguards): CTI will provide proactive data to our systems to meet regulatory requirements for risk analysis and information system activity review.

- SecureHealth Access Control & Password Policy: The knowledge that comes from the dark web of stolen credentials will directly activate enforcement measures in our Identity and Access Management (IAM) systems.
- SecureHealth Business Continuity & Disaster Recovery (BC/DR) Policy: The SecureHealth business continuity and disaster recovery (BC/DR) policy focuses on using threat intelligence to secure Veeam backup configuration against any ransomware variants threatening backups.

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2. Threat Intelligence Needs Assessment

Risk Assessment:
Conduct a risk assessment, detailing the most significant threats and vulnerabilities facing the organization.

Threat / Vulnerability	Likelihood	Impact	Description / Vector
Ransomware Attacks	High	High	Phishing emails or unpatched vulnerabilities can lead to network-wide encryption and halt patient care.
Unauthorized AccessHigh	High	High	Stolen or brute-forced corporate credentials are traded on dark web broker sites, breaching EMR systems.
Third-Party Vendor Risk	Medium	High	Compromise of an integrated telehealth vendor or IT supplier with trusted network access.

Stakeholder Collaboration:
Document feedback from stakeholders to determine the types of intelligence (tactical, operational, strategic) most beneficial to the organization.

- IT Security Team (Tactical/Operational Intelligence): Must be able to have immediate technical indicators (IP, File Hashes, Malicious Domain, etc.) and specific ransomware TTPs that can be used to optimize CrowdStrike EDR policies and firewall rules.
- Compliance Team (Strategic Intelligence): Needs to be aware of the high-level threat trends, post-mortems on regulatory breaches, and proof of being proactive about identifying threats.
- Incident Response Team (Operational/Tactical Intelligence): Needs actionable information when the attack is in progress, such as how the malware is behaving or how it is moving laterally, to help expedite the containment process.

Prioritization of Intelligence Needs:
Explain the rationale for prioritizing specific intelligence needs, focusing on high-impact and high-likelihood threats.

Rationale: Ransomware combined with stolen credentials presents the shortest path to catastrophic operational failure and severe regulatory penalties. Preventing an initial intrusion via stolen credentials directly mitigates the highest-likelihood ransomware deployment vector.

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3. Gathering and Integrating Intelligence Sources

Intelligence Sources:
List and describe the sources for threat intelligence, including internal sources (e.g., security logs, SIEM data) and external sources (e.g., threat feeds, OSINT, ISACs).

- Internal Sources: CrowdStrike EDR telemetry, Veeam backup access logs, corporate network security logs and Active Directory audit trail.
- External Sources (Commercial): Recorded Future subscription to real-time, analyzed technical threat feeds, profiles of threat actors and dark web credential monitoring.
- External sources (Open-Source/Community): Healthcare Information Sharing and Analysis Center (H-ISAC) feeds, Cybersecurity and Infrastructure Security Agency (CISA) alerts, trusted open source OSINT repositories (e.g., URLHaus).

Integration Process:
Document how these sources are integrated into the organization's existing security infrastructure, describing any tools or platforms used.

Centralized ingestion is achieved via our Threat Intelligence Platform (TIP), Recorded Future. The TIP automatically aggregates, deduplicates, and correlates external feeds with internal telemetry. Validated, high-confidence indicators are then pushed via API directly to CrowdStrike EDR to dynamically update blocklists and detection rules, ensuring our perimeter defenses adapt without manual security team intervention.

Source Validation:
Explain the processes for validating the credibility and relevance of each intelligence source.

- Prior to triggering automated blocks at a high severity level, a Data Cross-Referencing process must be completed between the IOC received from an open-source feed and the Recorded Future commercial data and H-ISAC community consensus.
- Relevance Scoring: Threats are automatically filtered out if they target operating systems and/or software suites that are not included in the active asset inventory of SecureHealth.

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4. Development of Threat Intelligence Processes

Data Collection Procedures:
Describe the methods for collecting raw threat data and any specific techniques used (e.g., Indicators of Compromise).

Threat intelligence is constantly collected and retrieved from sources of raw content via automated API pulls to Recorded Future. This encompasses a collection of atomic indicators (IPs, hashes) as well as behavior (registry changes, unusual PowerShell arguments) that are directly tied to the MITRE ATT&CK framework.

Analysis and Insight Generation:
Outline the analysis techniques employed to derive actionable insights, focusing on identifying patterns, attack tactics, or emerging threats.

- The Cyber Kill Chain & MITRE ATT&CK Mapping: Incoming ransomware threats are broken down and identified as which step of the attack they are in (Initial Access vs. Lateral Movement).
- Trend Analysis: Reviewing historical log data and Recorded Future profiles of ransomware groups to identify if a particular ransomware group is targeting mid-sized healthcare providers.

Dissemination Protocols:
Detail the protocols for sharing intelligence insights with relevant teams (e.g., security operations, incident response).

- High-Priority/Critical Threats (e.g., Active Ransomware Outbreak): Automated alerts that are sent in real-time to the Incident Response Team through specific PagerDuty or emergency Slack channels by Recorded Future.
- Medium Priority Threats (e.g., new TTPs discovered): Sent individually to the IT Security and Compliance teams as a weekly email threat briefing for defensive modifications.
- Strategic Threats (e.g., Shifts in Healthcare Regulations/Global Threat Landscape): Summarized in a Quarterly CTI Briefing presented to executive leadership and Compliance Board.

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5. CTI Capability and Infrastructure Building

Tool and Platform Acquisition:
List the tools and platforms (e.g., Threat Intelligence Platforms) acquired or developed to support CTI efforts.

- Recorded Future (TIP): Is the brain for external intelligence collection, correlation and analysis.
- CrowdStrike (EDR): Is the main operational enforcement mechanism that eats tactical intel to protect endpoints.
- Veeam: Used for data protection, using intelligence of known ransomware extension profiles to identify suspicious patterns of ransomware encryption early.

Team Development and Training:
Describe the personnel structure for CTI operations, including any specific roles, responsibilities, and training requirements.

- CTI Lead Analyst: Responsible for overseeing Recorded Future management, executing advanced analysis, and authoring strategic reports.
- Security Operations Center (SOC) Security Analysts: Tasked with daily triage of alerts generated by tactical intelligence matches.
- Required Training: Analysts must maintain certifications such as *GIAC Cyber Threat Intelligence (GCTI)* and complete specialized training modules provided by CrowdStrike and Recorded Future.

Capability Evaluation:
Document the organization's current CTI capabilities, identifying any gaps and future investment needs.

- Current Capabilities: Advanced capability for the ingestion of technical indicators through automation processes; highly mature endpoint blocking capabilities using CrowdStrike EDR.
- Gaps: Insufficient capacity internally to do malware reverse engineering manually; slight lack of visibility in terms of third-party vendor infrastructure risk exposure.
- Investments Needed: Implementation of a SOAR platform to further enable playbook automation based on intelligence gathered involving compromised employee credentials.

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6. Communication and Collaboration Channels

Internal Communication Channels:
List established channels for communicating CTI insights within the organization (e.g., with IT, management, security teams).

- SOC to IR Team: Integration of CrowdStrike alarms to the ticket management system (such as Jira Service Desk) for immediate escalation.
- Security to Compliance: Regular synchronization meeting to show blocked threat vectors, showing compliance with HIPAA security requirements.

- Security to Executive Leadership: Dashboard reports to show risks related to the organization's threat landscape.

External Collaboration:
Document partnerships with industry peers, law enforcement, or intelligence-sharing communities, specifying how these relationships enhance threat awareness.

- H-ISAC Membership: Regular sharing of de-identified local observations of indicators of bad actor activity with the Healthcare ISAC organization.
- CISA and FBI InfraGard: Attendance at regional briefings to ensure awareness of nation-state or widespread cybercrime campaigns against healthcare facilities.

Collaboration Challenges:
Discuss any challenges faced in establishing collaboration channels and strategies for improvement.

- Challenge: The delay caused by internal legal review processes makes it difficult for us to exchange threat data externally because there is a risk of exposing patient data.
- Mitigation Strategy: Develop sanitized playbooks that automatically remove any network internal identifiers and patient data before sharing technical indicators with H-ISAC.

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7. CTI Effectiveness Measurement and Optimization

Key Performance Indicators (KPIs):
Define the metrics used to measure CTI effectiveness (e.g., reduced response time, detection rate improvements).

- Mean Time to Detect (MTTD): Aim to reduce the time needed for detecting the presence of ransomware on endpoint systems by 30% with help from predictive Recorded Future indicators.
- False Positive Rate: Reduce the False Positive Rate to below 5% using a stringent Admiralty Scale validation process.
- Efficiency of Patching Priority Process: Measuring the efficiency in which IT patch vulnerabilities listed as "Exploited in the Wild" by Recorded Future compared to conventional CVSS scores.

Continuous Improvement Process:
Outline the feedback and review mechanisms in place to adapt the CTI strategy to new threats and evolving risks.

- Post-Incident Reviews: Following any security incident or major near-miss, a comprehensive review will evaluate whether the CTI strategy successfully predicted or flagged the threat TTPs.

- Semi-Annual Playbook Tuning: Detection logic and integration parameters within CrowdStrike and Recorded Future will be modified twice a year to adjust to evolving ransomware delivery methods.

Stakeholder Feedback:
Describe how feedback from stakeholders is gathered, documented, and applied to optimize the CTI strategy.

- Feedback Loops: Each quarter will close out with the distribution of an official survey to the teams of IT Security, Compliance, and IR.
- Application: In the event the IR team finds the threat briefings to be too long or lacking any practical use, the CTI team will shift the structure toward concise bullet points with an emphasis on action.